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研究方向

地球气候系统中陆冰的物理机制及其与大气、海洋之间的相互作用；南极和格陵兰冰盖不稳定性及其对海平面上升的影响；青藏高原冰川与水资源变化、冰川灾害；物理模式、人工智能的融合；自主研发国内首个二维（PoLIM）和三维（PoLarIS）陆冰动力模式；曾全程参加 IPCC AR6 中的南极冰盖模式比较计划，目前正在参加 IPCC AR7 冰盖模式比较计划 ISMIP7。

教育背景

中国科学院寒区旱区环境与工程研究所，自然地理，理学博士	2010–2013
中国科学院寒区旱区环境与工程研究所，自然地理，理学硕士	2007–2010
西北工业大学，材料物理，理学学士	2003–2007

工作经历

北京师范大学，地理科学学部，副教授	2021 至今
美国洛斯阿拉莫斯国家实验室，固体力学与流体动力学研究组，博士后	2017–2020
中国气象科学研究院，气候系统研究所，助理研究员	2015–2017
美国南卡罗来纳大学，数学系，博士后	2013–2015

科研奖项

国际冰川学会 Richardson 奖（ISMIP6 团队）	2022
施雅风冰冻圈与环境基金青年科学家奖	2023
清华大学-浪潮集团计算地球科学青年人才奖	2017

主要项目

国家重点研发项目“气候反转点的机制、影响及应对”(研究骨干)	2023–2028
基金委面上项目“冰架损伤对南极冰盖动力过程的影响研究”(主持)	2023–2026
国家重点研发项目“格陵兰冰盖监测、模拟及气候影响研究”(研究骨干)	2023–2026
新疆三次科考项目“新疆冰冻圈功能服务”(研究骨干)	2022–2025
中科院青藏所地球系统全重自主课题“冰崩过程数值模拟”(主持)	2022–2024
北师大地表国重自主课题“喜马拉雅山冰川与水资源变化模拟”(主持)	2021–2023
基金委青年项目“三维 Stokes 冰流模型模拟东南极冰川热力学特征”(主持)	2017–2019

社会服务

IPCC AR7 评估报告贡献作者	2026 至今
国际数字地球学会中委会数字极地专委会常委	2026 至今
第五次气候变化国家评估报告贡献作者	2025 至今
中国气象学会冰冻圈与极地气象专业委员会理事	2025 至今
Solid Earth under Greenland coordination committee 成员	2025 至今
《极地研究》编委	2025 至今
中国冰冻圈科学学会理事	2025 至今
组织开设“陆冰模式暑期班”	2024
极地青年科学家协会中国委员会副主席	2023–2025
中国第四纪科学研究会会员	2022-至今
Frontiers 期刊客座编辑	2018–2020

陆冰模式

自主模式 Polar Land Ice Simulator (PoLarIS)	主导开发
自主模式 Polythermal Land Ice Model (PoLIM)	主导开发
美国能源部陆冰模式 MPAS-Albany Land Ice Model (MALI)	参与开发

野外考察

祁连山八一冰川考察	2026
新疆慕士塔格冰川考察	2025
藏东南冰川考察	2024
第 40 次中国南极内陆考察	2023–2024
第 7 次中国北极考察	2016
格陵兰 NEEM 深冰芯考察	2010
喜马拉雅山珠峰和雅江源冰川考察	2008–2011

国际会议

国际冰盖模式比较计划（ISMIP7）会议，丹麦哥本哈根	2026
冰冻圈与气候（CliC）会议，新西兰惠灵顿	2026
美国地球物理年会（AGU），美国新奥尔良	2025
国际大气-海洋-冰冻圈联合大会，韩国釜山	2025
冰冻圈与气候变化国际研讨会，中国北京	2024
美国能源部地球系统模式年会，美国华盛顿	2020
美国地球物理年会（AGU），美国旧金山	2019

主要论文

- Gao, F., Shen, Q., Wang, H., **Zhang, T.***, Jiang, L., Liu, Y., Shum, C. K., An, Y., and Zhang, X.: Investigating the impact of sub-ice shelf melt on Antarctic ice sheet spin-up and projections, The Cryosphere, 20, 1947–1965, https://doi.org/10.5194/tc-20-1947-2026, 2026.
- **Zhang, T.***, Yang, W., Wang, Y., Zhao, C., Long, Q., Xiao, C. Numerical modeling of ice detachment tipping processes: insights from the Sedongpu Glacier, southeastern Tibetan Plateau, The Cryosphere, 19, 4487–4498, 2025.
- **Zhang, T.***, Wang, Y., Leng, W., Zhao, H., Colgan, W., Wang, C., Ding, M., Sun, W., Yang, W., Li, X., Ren, J. and Xiao, C. Projections of peak water timing from the East Rongbuk Glacier, Mt. Everest, using a higher order ice flow model. Earth's Future, 12(5), 2024.
- **Zhang, T.***, Colgan, W., Wansing, A., Løkkegaard, A., Leguy, G., Lipscomb, W.H. and Xiao, C. Evaluating different geothermal heat-flow maps as basal boundary conditions during spin-up of the Greenland ice sheet. The Cryosphere, 18(1), 387–402, 2024.
- Xiao, C.D., **Zhang, T.**, Che, T., Wei, Z.Q., Wu, T.H., Huang, L., Ding, M.H., Liu, Q., Wang, F.T., Wang, P.L. and Chen, J. Do abrupt cryosphere events in High Mountain Asia indicate earlier tipping point than expected?. Advances in Climate Change Research, 2023.
- Seroussi, H. and others including **Zhang, T.**: Insights into the vulnerability of Antarctic glaciers from the ISMIP6 ice sheet model ensemble and associated uncertainty, The Cryosphere, 17, 5197–5217， 2023.
- Zhao, H., Su, B., Lei, H., **Zhang, T.***, and Xiao, C. A new projection for glacier mass and runoff changes over High Mountain Asia. Science Bulletin, 2023.
- Yan, Z., **Zhang, T.***, Wang, Y., Leng, W., Ding, M., Zhang, D., and Xiao, C. Dynamic Evolution Modeling of a Lake-Terminating Glacier in the Western Himalayas Using a Two-Dimensional Higher-Order Flowline Model. Remote Sensing, 14(24), 6189, 2022.
- Yan, Z., Leng, W., Wang, Y., Xiao, C., **T. Zhang***. A comparison between three-dimensional, transient, thermomechanically coupled first-order and Stokes ice flow models. Journal of Glaciology, 1-12, 2022.
- Payne A J, Nowicki S, Abe Ouchi A, et al including **T. Zhang**. Future sea level change under coupled model intercomparison project phase 5 and phase 6 scenarios from the Greenland and Antarctic ice sheets. Geophysical Research Letters, 48(16): e2020GL091741, 2021.
- Edwards, T. and others including **T. Zhang**. Quantifying uncertainties in the land ice contribution to sea level rise this century, Nature, 593, 74–82, 2021.

- Seroussi, H. and others including **T. Zhang**. ISMIP6 Antarctica: a multi-model ensemble of the Antarctic ice sheet evolution over the 21st century, The Cryosphere, 14, 3033–3070, 2020.
- Sun, S. and others including **T. Zhang**. Antarctic ice sheet response to sudden and sustained ice shelf collapse (ABUMIP), Journal of Glaciology, 1–14, 2020.
- **Zhang, T.**, S. Price, M. Hoffman, M. Perego, X. Asay-Davis. Diagnosing the sensitivity of grounding line flux to changes in sub-ice shelf melting. The Cryosphere, 14, 3407–3424, 2020.
- Wang, Y., **T. Zhang***, C. Xiao, J. Ren, Y. Wang. A two-dimensional, higher-order, enthalpy-based thermo-mechanical ice flow model for mountain glaciers and its benchmark experiments. Computer & Geosciences, 141, 104526, 2020.
- Levermann, A. and others including **T. Zhang**. Projecting Antarctica's contribution to future sea level rise from basal ice-shelf melt using linear response functions of 16 ice sheet models (LARMIP-2), Earth System Dynamics, 11(1), 35–76, 2020.
- Seroussi, H. and others including **Zhang, T.**: initMIP-Antarctica: An ice sheet model initialization experiment of ISMIP6, The Cryosphere, https://doi.org/10.5194/tc-2018-271, 2019.
- Hoffman, M. J., Perego, M., Price, S. F., Lipscomb, W. H., **T. Zhang**, Jacobsen, D., Tezaur, I., Salinger, A. G., Tuminaro, R., Bertagna, L. MPAS-Albany Land Ice (MALI): A variable resolution ice sheet model for Earth system modeling using Voronoi grids, Geosci. Model Dev., https://doi.org/10.5194/gmd-2018-78, 2018.
- Y. Wang, **T. Zhang***, J. Ren, X. Qin, Y. Liu, W. Sun, J. Chen, M. Ding, W. Du, and D. Qin. An investigation of the thermo-mechanical features of Laohugou Glacier No.12 in Mt. Qilian Shan, western China, using a two-dimensional first-order flow-band ice flow model, The Cryosphere, doi:10.5194/tc-2016-38, 2018.
- **Zhang, T.**, S. Price, L. Ju, W. Leng, J. Brondex, G. Durand, and O. Gagliardini. A comparison of two Stokes ice sheet models applied to the Marine Ice Sheet Model Intercomparison Project for plan view models (MISMIP3d), The Cryosphere, 11, 179–190, 2017.
- **Zhang, T.***, M. Ding, C. Xiao, D. Zhang, Z. Du, Temperate ice layer found in the upper area of Jima Yangzong Glacier, the headstream of Yarlung Zangbo River, Science Bulletin, 61(8), 619–621, 2016.
- **Zhang, T.**, L. Ju, W. Leng, S. Price and M. Gunzberger, Thermomechancially coupled modelling for land-terminating glaciers: a comparison of two-Dimensional, first-order and three-dimensional, full-Stokes approaches. Journal of Glaciology, 61(227), doi: 10.3189/2015JoG14J220, 2015.
- **Zhang, T.**, C. Xiao, W. Colgan, X. Qin, W. Du, X. Liu, W. Sun, Y. Liu and M. Ding. Observed and Modelled ice temperature and velocity along the main flowline of East Rongbuk Glacier, Qomolangma (Mount Everest). Journal of Glaciology. 59(215), 438–448, doi:10.3189/2013JoG12J202, 2013.